

MILESTONE 04 / DELIVERY RECORD

Sandbox execution

Implemented; runtime blocked

Rootless Docker controller, immutable guard, offline commands, streaming output and enforced resource checks.

What was built

Rootless Docker controller, immutable guard, offline commands, streaming output and enforced resource checks.

Debugging and evidence

12 Java sandbox policy/lifecycle tests plus 17 Python guard tests pass using controlled fixtures/fake transport.

Remaining limitations

Actual image build and container execution not run; Docker daemon permission denied. Offline dependency cache is intentionally narrow.

Scope of this document

The following pages list the 8 source, configuration, test and documentation files assigned to this stage, with their purpose. Shared files are assigned to their principal delivery stage; the complete guide lists every authored file. Generated dependencies, caches, private settings and saved repository data are excluded.

Owner workflow

All development stages were authorized together. In-product approval is still required before repairs and again before publishing. No remote GitHub changes were made during development verification.

Folder structure

```
backend/  
  src/  
    main/  
      java/  
        dev/  
          codebasedoctor/  
            sandbox/  
              DockerSandbox.java  
              SandboxPolicy.java  
        test/  
          java/  
            dev/  
              codebasedoctor/  
                sandbox/  
                  DockerSandboxTest.java  
                  SandboxPolicyTest.java  
  docker/  
    guard.py  
    sandbox.Dockerfile  
    test_guard.py  
  scripts/  
    preflight.sh
```

Files and responsibilities

`backend/src/main/java/dev/codebasedoctor/sandbox/DockerSandbox.java`

Updated from foundation | Delivery stage 4

Checks engine/image readiness, launches isolated offline containers, streams bounded output, verifies reports and cleans up after failures/cancellation.

`backend/src/main/java/dev/codebasedoctor/sandbox/SandboxPolicy.java`

Created | Delivery stage 4

Validates sandbox jobs, immutable image IDs, source import payloads and fixed execution policy.

`backend/src/test/java/dev/codebasedoctor/sandbox/DockerSandboxTest.java`

Created | Delivery stage 4

Nine fake-transport tests exercise readiness, isolation flags, results, streaming and container lifecycle behavior without executing Docker.

`backend/src/test/java/dev/codebasedoctor/sandbox/SandboxPolicyTest.java`

Created | Delivery stage 4

Three tests enforce bounded imports, source-path safety and execution policy.

`docker/guard.py`

Created | Delivery stage 4

Immutable container helper: checks enforced limits, imports bounded files, invokes fixed offline builds, quiesces build processes and parses real JUnit reports.

`docker/sandbox.Dockerfile`

Created | Delivery stage 4

Builds the trusted Java/Maven/Gradle execution image with independent non-root helper/build users and a reviewed offline dependency seed.

`docker/test_guard.py`

Created | Delivery stage 4

Seventeen Python tests for cgroup checks, archive import bounds, safe JUnit parsing, offline commands and process/report safety.

`scripts/preflight.sh`

Created | Delivery stage 4

Read-only Docker prerequisite checks; never installs services or changes socket permissions.